

Identifying the Performance Descriptor in Direct Syngas Conversion to Long-Chain α -Olefins over Ruthenium-Based Catalysts Promoted by Alkali Metals

Hailing Yu, Yao Wei, Tiejun Lin, Caiqi Wang, Yunlei An, Fei Yu, Fanfei Sun, Zheng Jiang, Yuhua Sun, and Liangshu Zhong*



Cite This: *ACS Catal.* 2023, 13, 3949–3959



Read Online

ACCESS |



Metrics & More



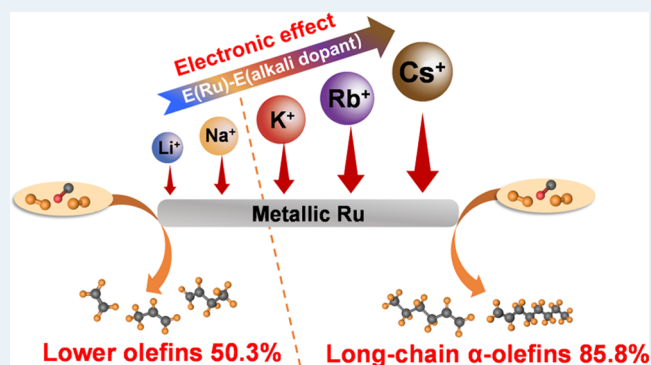
Article Recommendations



Supporting Information

ABSTRACT: Fischer-Tropsch synthesis (FTS) provides a non-petroleum route for the production of value-added α -olefins from syngas. However, regulating the product distribution to promote the formation of long-chain α -olefins while maintaining low selectivity to C1 byproducts (CO_2 and CH_4) remains a great challenge. Herein, a series of alkali metal-promoted Ru/SiO₂ catalysts were synthesized to investigate the effect of alkali metals and identify the performance descriptor for syngas to α -olefins. X-ray absorption and photoelectron spectroscopy as well as CO diffuse reflectance infrared Fourier transformation spectra revealed that Ru's electronic structure can be tuned by the addition of alkali metal promoters. The Allen scale electronegativity deviation between the metallic Ru and the alkali metal promoter, which was identified as a performance descriptor, was correlated with the initial turnover frequency (TOF) and olefins distribution, as well as the chain growth probability. In addition, the fraction of olefin products was successfully adjusted from lower olefins (50.3%) to long-chain α -olefins (85.8%) by controlling the kinds of alkali metal promoters. Evidences from X-ray absorption spectroscopy and chemisorption characterizations revealed that the controllable Ru electron density could enhance CO adsorption and hinder the reactivity of chemisorbed H species, thus benefiting the α -olefin production. These results offer a comprehensive view of the electronic effect for regulating product selectivity during the Fischer-Tropsch reaction process.

KEYWORDS: Fischer-Tropsch synthesis, syngas conversion, alkali metal, long-chain α -olefins, ruthenium-based catalyst



1. INTRODUCTION

Long-chain α -olefins (C_{5+}) are versatile industrial feedstocks for the production of a variety of value-added products such as synthetic lubricants, surfactants, new polymers, biodegradable detergents, high-octane gasoline, and agricultural chemicals.^{1–3} The prevalent method of α -olefin production is mainly based on oligomerization of ethylene, which is mostly derived from petroleum resources.^{4,5} With the rapid depletion of petroleum resources, direct transformation of nonpetroleum carbon resources into olefins has received great interest from the academic and industrial community.

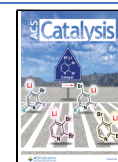
Fischer-Tropsch synthesis (FTS) provides an important platform technology to produce various fuels and chemicals from syngas (a mixture of H_2 and CO) derived from readily available carbon resources, such as coal, natural gas, biomass, shale gas, solid waste, and even CO_2 .^{6,7} The direct synthesis of olefins via FTS is also called Fischer-Tropsch to olefins (FTO).^{8,9} The selectivity of olefins with a certain range of carbon number is always limited by the Anderson-Schulz-Flory (ASF) product distribution due to the surface polymerization

mechanism, and the C1 byproduct (CH_4 and CO_2) yield is always relatively high (>30%), which decreases the carbon efficiency of the FTO technology.^{10,11} Recently, great progress has been achieved in the production of α -olefins over Fe-based and Co-based catalysts.^{12–15} Xu et al. presented a hydrophobic core-shell FeMn@Si catalyst with 68.2% α -olefin selectivity and less than 22.5% C1 byproduct selectivity.¹⁶ Gong et al. reported that graphene-like carbon-confined metallic Co (Co@C) exhibited an excellent performance for the FTO reaction with 56.4% olefin selectivity, and the CO_2 selectivity was also suppressed within 6%.¹⁷ Prieto et al. found that the modified Co/ γ - Al_2O_3 catalyst can achieve 19.9% selectivity to C_{5-10} olefins with negligible C1 byproduct selectivity (<10%).¹⁸ However, it

Received: December 13, 2022

Revised: February 28, 2023

Published: March 7, 2023



is still necessary to develop a new strategy to regulate the product distribution to achieve a high yield of long-chain α -olefins.

Utilization of alkali metal promoters is effective in improving the catalytic performance for FTS.^{19–22} Alkali metals, including Li, Na, K, Rb, and Cs, possess variant electron donation capacities with different electronegativities, and usually serve as electronic and structural promoters to adjust the catalytic performance. In previous studies, K or Na was added to Fe-based and Co-based FT catalysts to facilitate the carbonization of catalysts and restrain the secondary hydrogenation of olefins, resulting in low methane selectivity and high olefin selectivity.^{23,24} Zhai et al. suggested that the synergistic effect of Na and Zn, serving as electronic and structure promoters, respectively, could tailor the production of long-chain olefins.¹² Xie et al. used Na and its counter-ion S as electronic promoters to modulate the metallic Co and greatly improve selectivity to lower olefins.²⁵ In our previous work, the addition of Na could stabilize the formation of Co₂C nanoprisms with exposed facets of (101) and (020), which achieved a high selectivity to lower olefins with suppressed CH₄ selectivity.^{26,27} Especially, the sodium-promoted metallic ruthenium nanoparticles exhibited high selectivity to α -olefins with ultralow C1 byproduct selectivity.²⁸ However, the underlying role of alkali metal promoters for Ru-based FTO remains unclear and there still lacks a detailed descriptor for a quantitative study on the electronic effect of alkali metal promoters.

Herein, emphasis was placed on defining the effects of alkali metals on the specific activity and product distribution over Ru-based FTO catalysts. A series of Ru/SiO₂ catalysts with different kinds of alkali metal (Li, Na, K, Rb, and Cs) promoters were prepared and denoted as AM-Ru/SiO₂ (where AM = alkali metal). The kind of alkali promoter was elucidated as the key factor affecting the olefins product distribution during the FTO process over Ru/SiO₂ catalysts. The relationship between the Allen scale electronegativity of alkali metals and catalytic performance was also established.

2. RESULTS AND DISCUSSION

2.1. Structure Properties of AM-Ru/SiO₂ Catalysts.

Table 1 shows the textural properties of various AM-Ru/SiO₂

Table 1. Textural Properties of Various Catalysts Determined by N₂ Adsorption–Desorption

catalysts	BET surface area (m ² ·g ^{−1})	pore volume (cm ³ ·g ^{−1})	average pore size (nm)
Ru/SiO ₂	307	1.2	15.5
Li-Ru/SiO ₂	267	1.2	18.6
Na-Ru/SiO ₂	241	1.0	17.4
K-Ru/SiO ₂	221	1.0	18.0
Rb-Ru/SiO ₂	221	1.0	19.1
Cs-Ru/SiO ₂	210	1.0	19.4

catalysts. The addition of alkali metal significantly reduced the specific BET surface area of AM-Ru/SiO₂ catalysts while retaining a similar pore volume and average pore size. The mesoporous structure of the SiO₂ support was maintained after the introduction of the alkali metal promoter (Figure 1a). The crystal structure of AM-Ru/SiO₂ catalysts was characterized by X-ray diffraction (XRD) (Figure 1b,c). The bulk RuO₂ (JCPDS 43-1027) was reduced to metallic Ru phase (JCPDS 06-0663) during the reduction process, and metallic Ru remained after the

reaction (Figure 1d). The crystallite size of metallic Ru for various reduced and spent samples was calculated using Scherrer's equation as shown in Table 2. The crystallite size of metallic Ru distinctly decreased with the introduction of the alkali metal promoter from 9.9 to ~6.0 nm and was maintained at ~6.0 nm after the reaction for all promoted catalysts.

Representative TEM images of the reduced and spent catalysts are presented in Figures 2 and S1. The Ru nanoparticles with similar particle sizes (4.7–5.7 nm) were homogeneously dispersed on the SiO₂ support after the reaction. The average size of the metallic Ru decreased with the introduction of alkali metal, which agreed well with the results based on XRD characterization. Metal dispersion of Ru was further analyzed by H₂ and CO chemisorption experiments. The H₂ and CO uptake obviously increased with the addition of the alkali metal promoter. The metal dispersion calculated by H₂ and CO chemisorption of the promoted catalysts was greatly increased compared to the unpromoted case, which was consistent with the TEM-derived metal dispersion.²⁹ The results summarized in Table 2 clearly suggest that the alkali metal promoters could greatly promote the dispersion of metallic Ru.

H₂-TPR was measured to verify the reduction property of alkali metal-promoted catalysts (Figure 3). All catalysts exhibited only one peak at 100–200 °C, which could be assigned to the reduction of bulk RuO₂ to metallic Ru. It was clearly observed that the reduction peak at 134 °C shifted to higher temperatures when alkali metal promoters were added, and the reduction peak temperature increased gradually with the decrease of alkali metal electronegativity. Obviously, the addition of alkali metal promoter inhibited the RuO₂ reduction.

DRIFTS of CO chemisorption revealed the change of the surface electronic state of metallic Ru. As shown in Figure 4, the AM-Ru/SiO₂ displayed a typical linear-type CO adsorption peak in the range of 2000–2050 cm^{−1}, a twin-type CO adsorption peak at about 1950 cm^{−1}, and a bridge-type adsorption peak at 1700–1800 cm^{−1} on the metallic Ru surface at 50 °C.³⁰ Obviously, the peak intensity of twin-type CO adsorption increased with a decrease in electronegativity of the alkali metal promoters. A red shift of the linearly adsorbed CO peaks was observed for the samples with addition of the alkali metal promoter. Such a red shift phenomenon became obvious with a decrease in electronegativity of the alkali metal promoter. The red shift of the CO adsorption peak indicated that the alkali metal promoter as an electronic donor enriched the density of the electronic charge in the d orbitals protruding from the metal surface. This could lead to an increment in the degree of backdonation of the charge from these orbitals to the π^* antibonding orbitals of the chemisorbed CO.^{31,32} The red shift difference of CO linear adsorbed peaks further proved that different kinds of electron donor additives lead to different electron densities of metallic Ru, thereby reducing the amplitude of the carbonyl stretch vibration frequency.

The change of electron density of Ru with the addition of alkali metal promoters was also investigated by X-ray photoelectron spectroscopy (XPS). As shown in Figure S2, a peak shift to lower binding energy was observed, suggesting the electron transfer from alkali metal promoters to the metallic Ru.³³ Especially, an obvious shift in Ru 3p_{3/2} binding energy from 462.3 eV for Ru/SiO₂ to 461.0 eV for Cs-Ru/SiO₂ was found (Figure 5a). A more detailed structural information at atomic level can be obtained by extended X-ray absorption fine structure (EXAFS). The EXAFS spectra of various samples exhibited only one strong peak at ~2.4 Å aligned with the Ru foil,

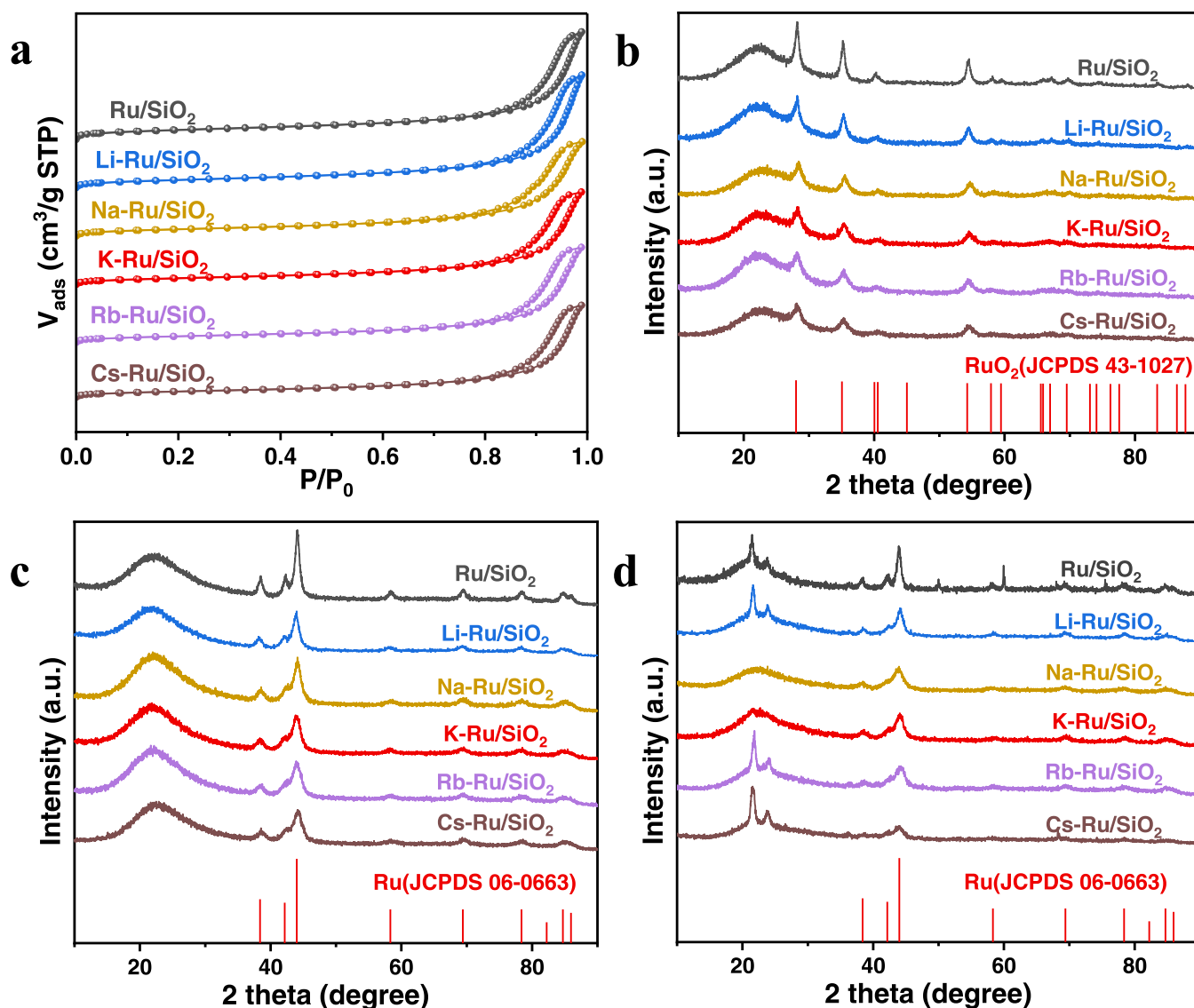


Figure 1. Structural characterization of various AM-Ru/SiO₂ catalysts. (a) N₂ adsorption–desorption isotherms; (b) XRD patterns for the fresh catalysts; (c) XRD patterns for the reduced catalysts; (d) XRD patterns for the spent catalysts.

Table 2. Ru Nanoparticle Size for Various Reduced and Spent Catalysts and Chemisorption Properties of Various Reduced Catalysts

catalysts	crystallite size (nm) ^a		particle size (nm) ^b		metal dispersion				
	reduced	spent	reduced	spent	<i>D</i> _{TEM} (%) ^c	H ₂ uptake (μmol·g ^{−1})	<i>D</i> _H (%) ^d	CO uptake (μmol·g ^{−1})	<i>D</i> _{CO} (%) ^e
Ru/SiO ₂	9.9	8.5	7.9	8.4	14.2	10.0	0.7	26.4	3.9
Li-Ru/SiO ₂	6.0	5.9	5.4	4.6	20.7	27.0	3.4	113.0	8.3
Na-Ru/SiO ₂	5.9	5.4	4.7	5.2	23.8	27.4	3.2	107.5	9.8
K-Ru/SiO ₂	6.1	5.8	5.4	5.2	21.5	23.5	2.8	122.1	8.9
Rb-Ru/SiO ₂	6.2	5.0	5.7	4.9	19.7	34.9	4.2	111.4	9.3
Cs-Ru/SiO ₂	5.6	5.8	5.3	5.3	21.1	31.1	4.2	117.2	8.8

^aCalculated using Scherrer's equation by XRD. ^bSurface-average Ru particle size distribution as determined by transmission electron microscopy (TEM). ^cMetal dispersion calculated by TEM. $D_{\text{TEM}} = 1.12/d_{\text{TEM}}$. ^dMetal dispersion calculated by H₂ chemisorption. ^eMetal dispersion calculated by CO chemisorption.

which was attributed to the Ru–Ru coordination (Figure 5b).³⁴ The corresponding FT EXAFS curve-fitting results are displayed in Table S1. The coordination number (CN) of Ru–Ru and the average interatomic distance (*R*) showed similar values, indicating that various samples possessed similar metallic Ru nanoparticle sizes.

The electronic structure of Ru atoms with different alkali metal promoters was further determined via X-ray absorption near-edge structure (XANES). As shown in Figure 5c, the near-edge adsorption energies (*E*₀) of all reduced samples are located close to the Ru foil and away from RuO₂, implying the unchanged metallic state of the Ru species. In addition, a distinct

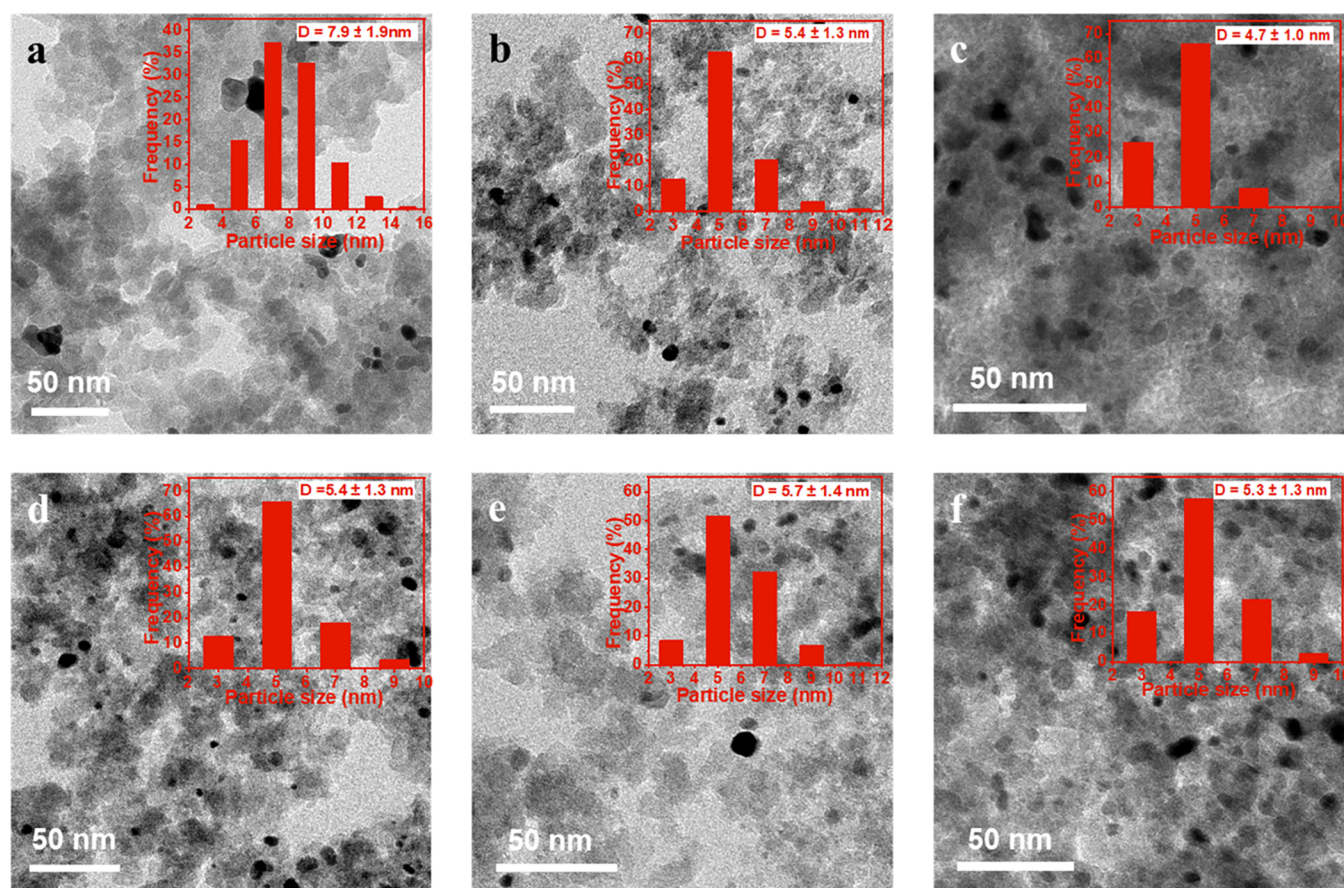


Figure 2. TEM images and the corresponding crystallite size distributions of various reduced catalysts. (a) Ru/SiO₂, (b) Li-Ru/SiO₂, (c) Na-Ru/SiO₂, (d) K-Ru/SiO₂, (e) Rb-Ru/SiO₂, and (f) Cs-Ru/SiO₂.

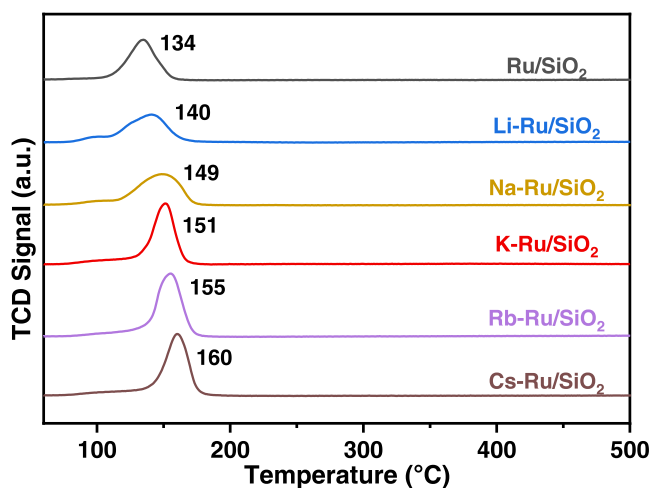


Figure 3. H₂-temperature-programmed reduction profiles for various alkali metal-promoted Ru/SiO₂ catalysts.

shift of the absorption edge toward a lower energy close to the Ru foil with the decrease of electronegativity of the alkali metal promoters is observed in the inset of Figure 5c. This suggests that the decrease of electronegativity of the alkali metal promoters leads to an increase in the electronic density of Ru atoms, as confirmed by the results of XPS, CO-DRIFTS, and XANES.³⁵ Therefore, the Allen scale electronegativity of alkali metals could be used as a descriptor to describe the effects of the electronic promoter quantitatively.

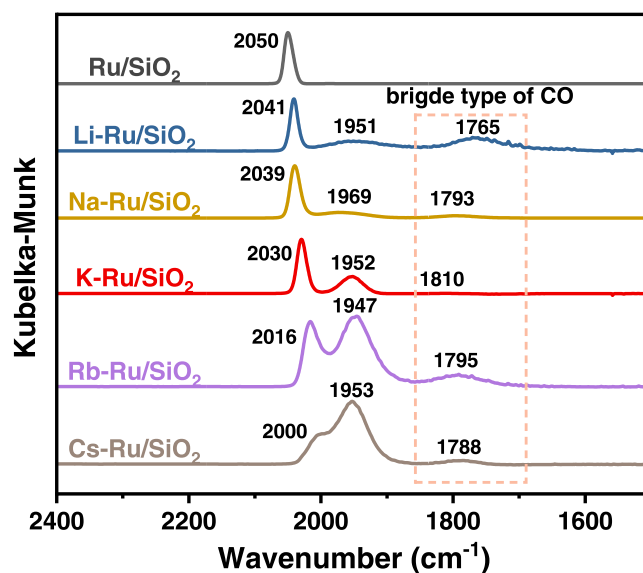


Figure 4. DRIFTS of CO chemisorption over AM-Ru/SiO₂ catalysts.

2.2. Catalytic Performance of AM-Ru/SiO₂ Catalysts.

The catalytic performance of various Ru-based catalysts was evaluated in a fixed-bed reactor. The bare Ru/SiO₂ catalyst exhibited a high CO conversion of 54.5% with low C1 byproduct selectivity. However, the paraffin selectivity reached 69.5% for Ru/SiO₂. In Figure 6, the Allen scale electronegativity difference between the metallic Ru and alkali metal (ΔEN) is used as a

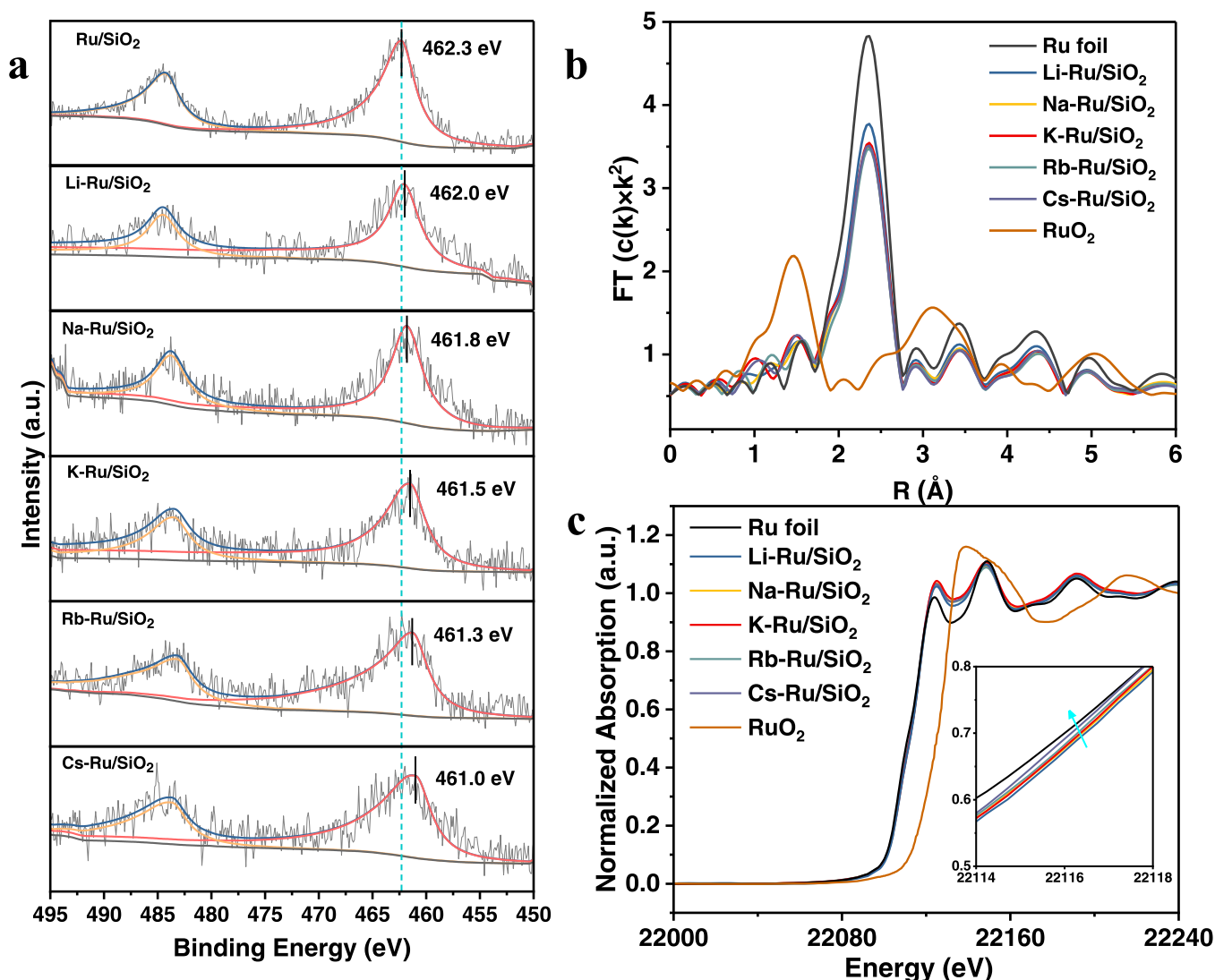


Figure 5. Electronic structure studies. (a) XPS analysis of various reduced AM-Ru/SiO₂ catalysts. (b) Ru K-edge EXAFS spectra of various reduced catalysts. (c) Ru K-edge X-ray absorption near-edge structure (XANES) spectra of various reduced catalysts.

descriptor to correlate the activity and product selectivity in the FTO reaction. The ΔEN value of the Ru/SiO₂ sample was defined as 0. Catalysts promoted by a high electronegative promoter showed a notably higher CO conversion (Figure 6a). The CO₂ selectivity was in the range of 0.2–7.0% for various samples and increased gradually with the decrease of alkali metal electronegativity (Figure 6b). The low CO₂ selectivity illustrated the low activity of water–gas shift reaction (WGS) over AM-Ru/SiO₂ catalysts. The slightly increased CO₂ selectivity may be attributed to the mildly enhanced basic sites of Rb-Ru/SiO₂ and Cs-Ru/SiO₂ catalysts. In addition, the CH₄ selectivity remained $\leq 4.2\%$ in all cases. Olefins were the dominant products among all of the alkali metal-promoted catalysts although olefin selectivity decreased gradually with the ΔEN value (Figure 6c). Compared to Ru/SiO₂, the addition of the alkali metal promoter led to the transformation of paraffin products to olefins. Notably, the differences in olefins' distribution showed a marked dependence on the Allen scale electronegativity of the alkali metal promoter. Catalysts promoted with a higher-electronegativity alkali metal (Li, Na) showed a higher selectivity to lower olefins (C_{2–4}); especially, 40.0% proportion of lower olefins was found for the Na-Ru/SiO₂

catalyst. However, the long-chain α -olefin (C₅₊) fraction increased when promoters with lower electronegativity (K, Rb, Cs) were applied. In addition, the C₅₊ α -olefin proportion was as high as 85.2% for the Cs-Ru/SiO₂ catalyst (Figure 6d).

As CO conversion may have a great effect on the product distribution, a comparative study was also performed under similar CO conversion levels by adjusting the space velocity.³⁶ In order to further explore the effect of alkali metal promoters on the olefin product distribution, the initial turnover frequency (TOF), product selectivity, and carbon number distribution were determined under similar CO conversions of 5–10% (Figure 7). The TOF value decreased linearly with decreasing ΔEN value from 0.106 to 0.052 s^{−1} (Figure 7a). With regard to the product selectivity, the addition of alkali metal led to high olefin selectivity, and the paraffin selectivity increased with the decrease of alkali metal electronegativity. It seemed that a higher-electronegativity alkali metal-promoted catalyst exhibited a higher C_{2–4} selectivity at similar CO conversion levels, while the decrease in the electronegativity of alkali metal promoter favored the formation of C₅₊ hydrocarbon products (Figure 7b,c). Li-Ru/SiO₂ and Na-Ru/SiO₂ catalysts displayed high C_{2–4} fractions (40.7 and 50.3%), while the alkali metal

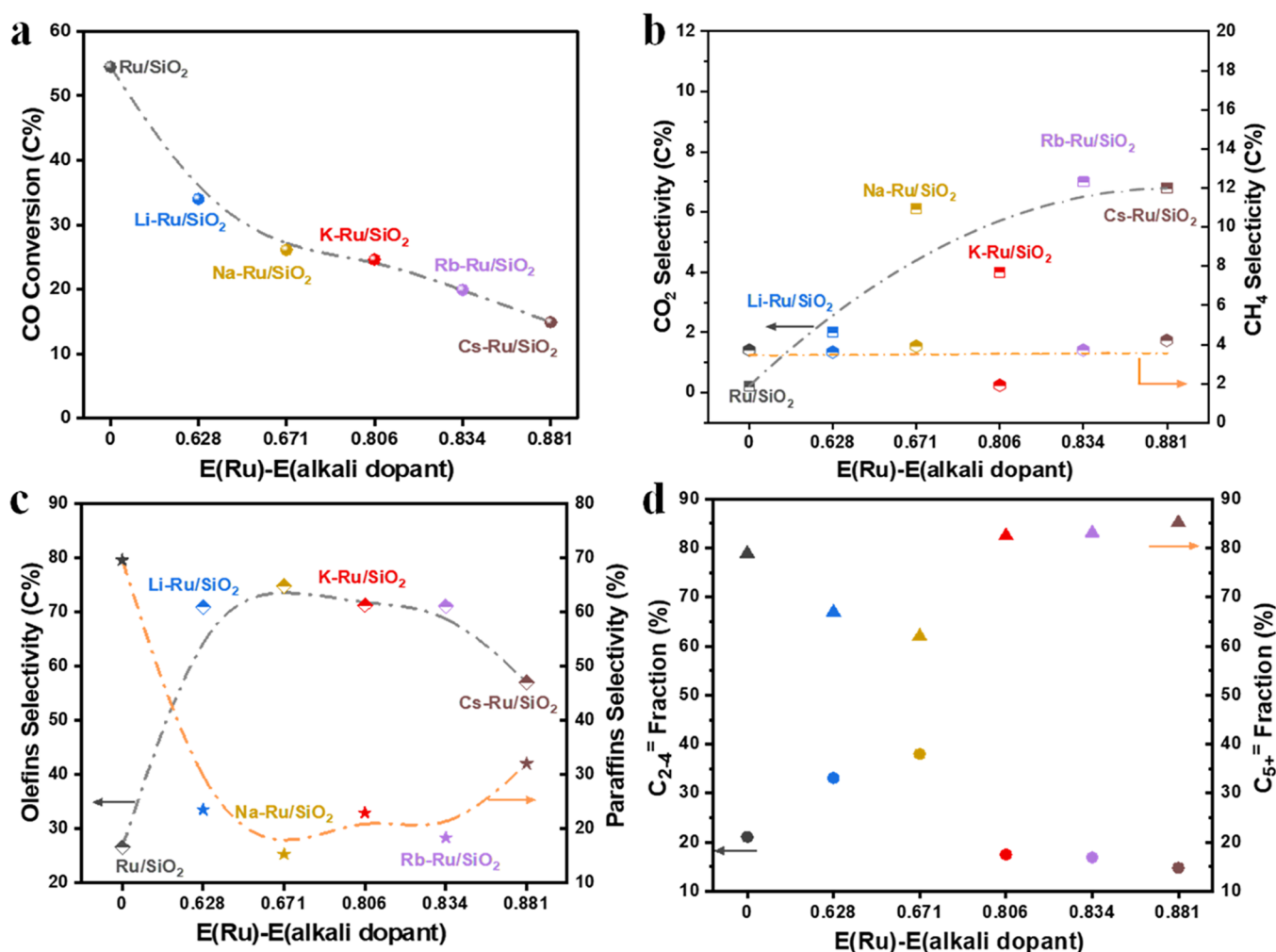


Figure 6. CO conversion (a), product selectivity (b, c), and olefin distribution (d) for AM-Ru/SiO₂ catalysts as a function of the Allen scale electronegativity difference between Ru and the alkali metal, represented by ΔEN . Reaction conditions: $T = 270^\circ\text{C}$, $P = 5$ bar, WHSV = 3000 mL·g_{cat}⁻¹·h⁻¹, H₂/CO = 2.

promoter with a lower electronegativity led to an increase of C₅₊ fraction (82.5, 83.1, and 85.2% for K-Ru/SiO₂, Rb-Ru/SiO₂, and Cs-Ru/SiO₂, respectively). Additionally, the selectivity to C₅₊ hydrocarbons increased from 79.9% for Ru/SiO₂ to 92.5% for Cs-Ru/SiO₂ with decreasing alkali metal promoter electronegativity, and an increase in chain growth probability (α) of the olefin product from 0.60 to 0.81 was also observed (Figure 7d). The olefin-to-paraffin (O/P) ratio with carbon number for various catalysts was also investigated, as shown in Figure S3. Obviously, Li-Ru/SiO₂ and Na-Ru/SiO₂ catalysts displayed much higher O/P values in the C₃–C₁₁ range. For hydrocarbons with longer carbon chains ($n > 11$), the O/P values over K-, Rb-, and Cs-promoted Ru/SiO₂ catalysts turned to be higher than those of Li-Ru/SiO₂ and Na-Ru/SiO₂.

As the catalysts promoted by K exhibited the highest yield of C₅₊ olefins fraction, the effect of K/Ru molar ratio on the catalytic performance was investigated. The catalyst without the K promoter displayed a high CO conversion (54.7%) with low olefin selectivity (26.6%). On increasing the K content, the CO conversion decreased significantly, while the olefin selectivity increased remarkably (Figure 8a). However, as reported in a previous work, excessive alkali metal additives would suppress the olefin production and promote the C1 byproduct formation.²⁸ The CO₂ selectivity gradually increased from 0.2

to 7.4% on increasing the K/Ru molar ratio. The CH₄ selectivity was maintained at about 2.0% for samples with K/Ru molar ratio in the range of 0.1–0.5, while it increased to 5.5% for the 1K-Ru/SiO₂ catalyst (Figure 8b). In addition, the chain growth probability decreased slightly on increasing the K/Ru molar ratio (Figure S4).

2.3. Structure–Performance Relationship Discussion.

The effect of alkali metals on the catalytic performance of Ru/SiO₂ catalysts was investigated in detail. As shown in Figures 6a and 7a, the catalytic activity based on CO conversion or TOF decreased with the addition of alkali metal promoters. In order to understand the effect of the alkali metal, the Ru 3p_{3/2} binding energy and wavenumber of linear-type CO were correlated with ΔEN (Figure 9a). The increase in electron density of the metallic Ru enhanced the extent of backdonation, which increased the strength of the carbon–metal bond and decreased the strength of the carbon–oxygen bond.³⁷

The uptakes of H₂ or CO on various reduced catalysts (pretreated at 450 °C for 2 h) are summarized in Table 2. The amounts of chemisorption are expressed by H/Ru or CO/Ru, which is the number of hydrogen atoms or CO molecules chemisorbed divided by the number of Ru atoms measured by ICP-OES (Table S2). Figure 9b and Table S2 show the amount of H₂ chemisorption after CO preadsorption. Obviously, the

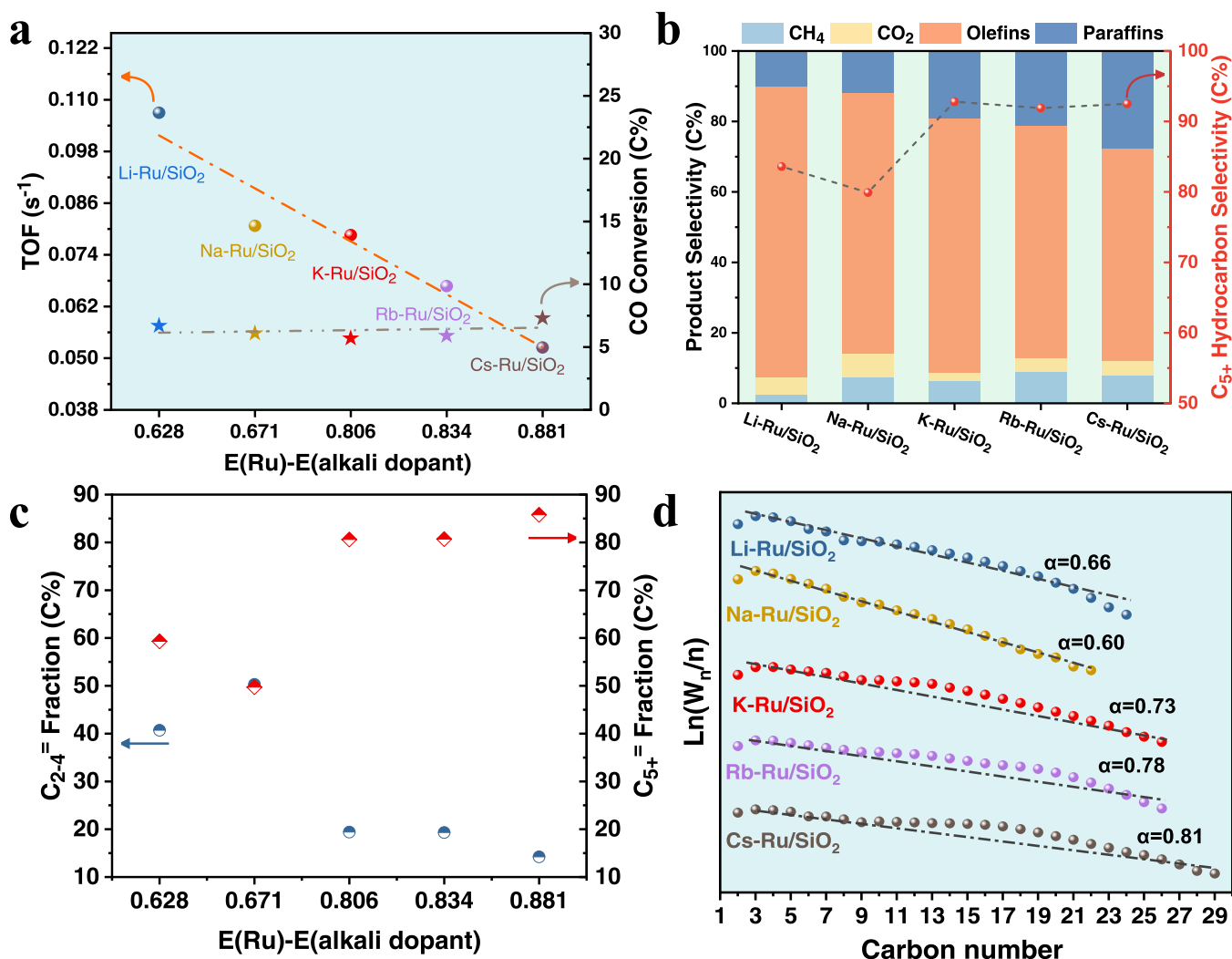


Figure 7. (a) Turnover frequency (TOF, per unit surface metal site as quantified with CO chemisorption) and CO conversion, (b) product selectivity and C₅₊ hydrocarbon selectivity, (c) olefins distribution including C₂₋₄ and C₅₊, and (d) ASF distribution for the olefin product obtained with various AM-Ru/SiO₂ catalysts under similar CO conversion levels of 5–10%. Reaction conditions: $T = 270\text{ }^{\circ}\text{C}$, $P = 5\text{ bar}$, $\text{WHSV} = 6000\text{--}12000\text{ mL}\cdot\text{g}_{\text{cat}}^{-1}\cdot\text{h}^{-1}$, $\text{H}_2/\text{CO} = 2$.

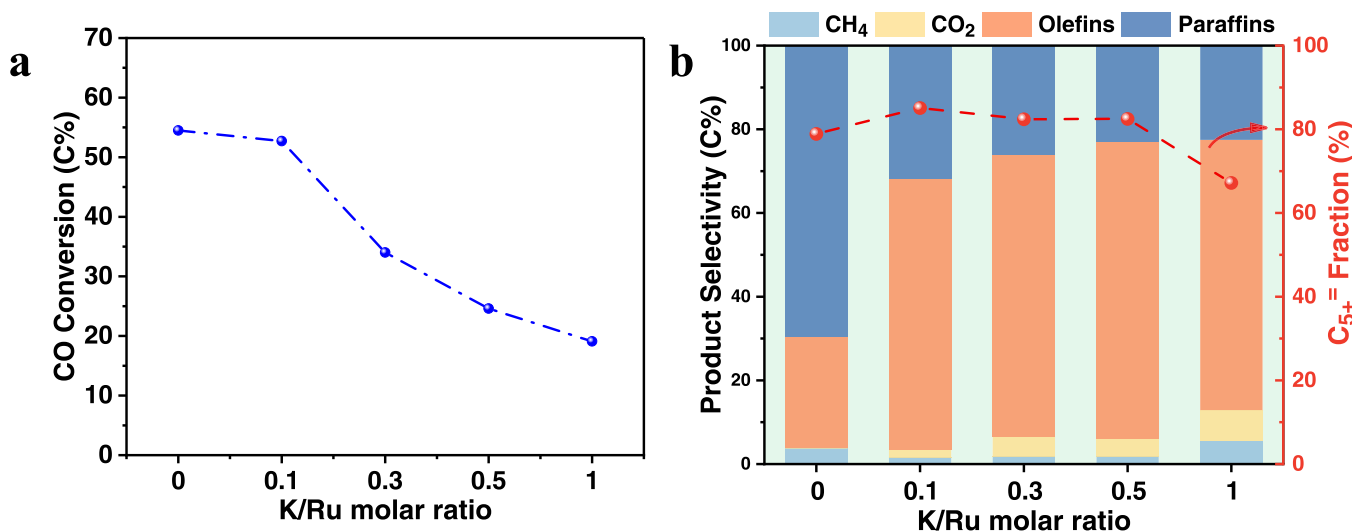


Figure 8. Catalytic performance of xK-Ru/SiO₂ ($x = 0, 0.1, 0.3, 0.5$, and 1.0) catalysts. (a) Effect of K/Ru molar ratio on CO conversion. (b) Product selectivity as a function of K/Ru molar ratio. Reaction conditions: $T = 270\text{ }^{\circ}\text{C}$, $P = 5\text{ bar}$, $\text{WHSV} = 3000\text{ mL}\cdot\text{g}_{\text{cat}}^{-1}\cdot\text{h}^{-1}$, $\text{H}_2/\text{CO} = 2$.

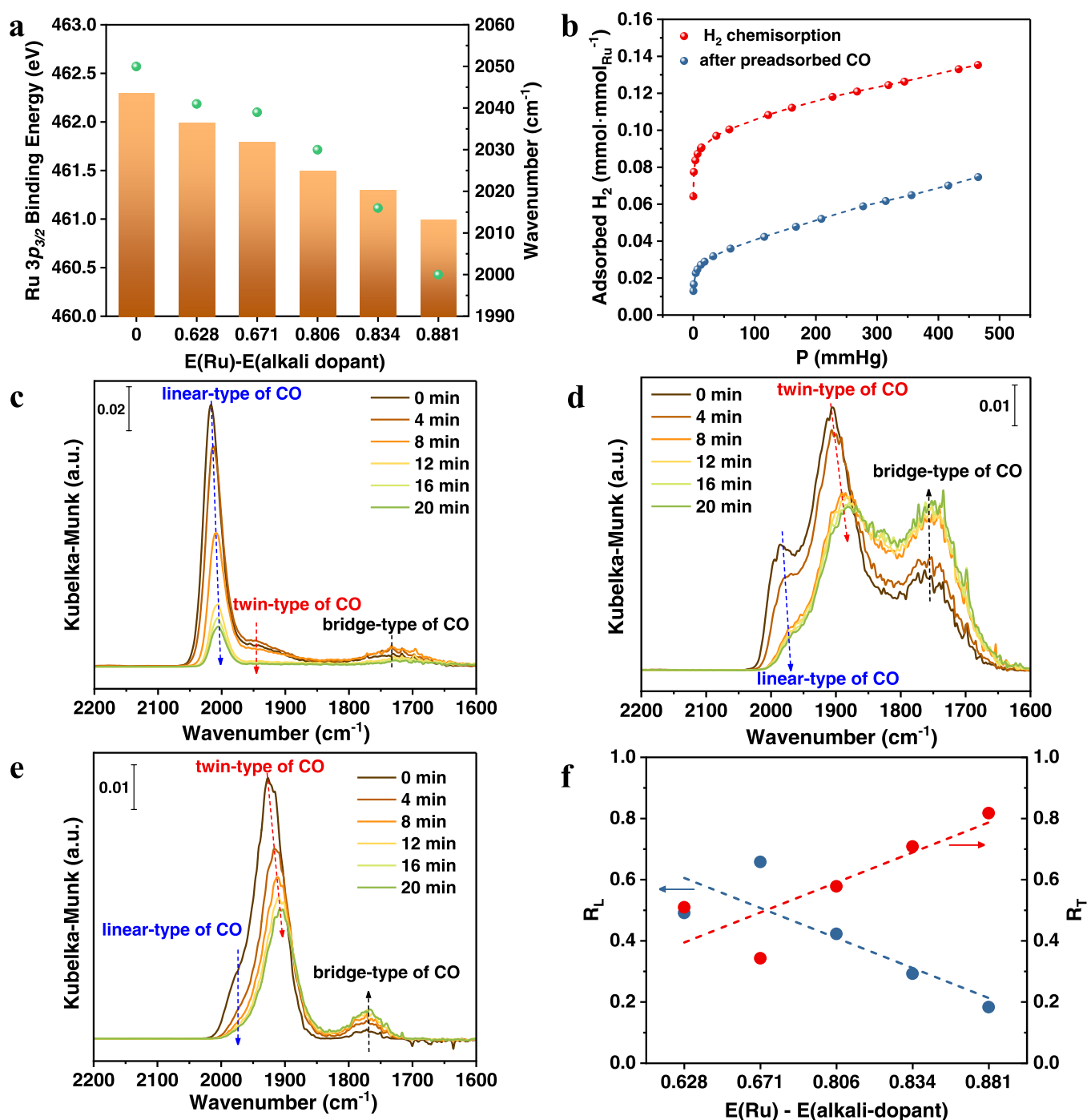


Figure 9. (a) Ru $3p_{3/2}$ binding energy and linear-type adsorbed CO wavenumber versus the difference in the Allen scale electronegativity of Ru and alkali metals (ΔEN). (b) Isotherms of H_2 chemisorption before and after CO preadsorption over K-Ru/SiO₂. Evolution of different types of adsorbed CO under H_2 flow over (c) Li-Ru/SiO₂, (d) K-Ru/SiO₂, and (e) Cs-Ru/SiO₂. (f) Relationship between the relative amount of linear and twin-type adsorbed CO and the Allen scale electronegativity difference between Ru and the alkali metal (ΔEN).

reduced catalysts can adsorb more H_2 than that with preadsorption of CO. It is reasonable to speculate that the adsorbed CO acts as a competitive component for the dissociative adsorption of H_2 . Accordingly, the adsorption strength of CO would affect the ability of hydrogenation and CO conversion. In situ CO-DRIFTS was applied to investigate the hydrogenation capacity of chemisorbed CO under the reaction temperature of 270 °C. Figure 9c–e depicts the CO-DRIFTS spectra after purging H_2 into CO-saturated catalysts. A characteristic frequency at 3015 or 3016 cm^{-1} was detected,

accompanied by the consumption of adsorbed CO, which was ascribed to the CH_x species (Figure S5).²⁹ It is worthy to note that the decreased intensity of linear-type CO is related to the appearance of CH_x , and this phenomenon was more remarkable on Li-Ru/SiO₂ than on K-Ru/SiO₂ and Cs-Ru/SiO₂, which was responsible for its higher activity in FTO reaction.

Comparison of the spectra in Figure 9c–e reveals the reactivity of adsorbed CO toward hydrogenation. The twin-type CO was more stable than the linear-type CO after reaction with H_2 , indicating that the twin-type CO was inactive for

hydrogenation. In order to compare the relative concentrations of the active linear-type and inactive twin-type CO, the infrared spectra of CO adsorbed recorded at 50 °C in the range of 2100–1850 cm^{-1} were zoomed in as shown in Figure S6. The ordinate of Figure 9f shows the integrated intensity of bands normalized by that of two bands in the range of 2050–1950 cm^{-1} obtained at 50 °C. The relative intensity of the linear adsorbed CO (R_L) and twin-type adsorbed CO (R_T) can be expressed as the following equations

$$R_L = A_L / (A_L + A_T) \quad (1)$$

$$R_T = A_T / (A_L + A_T) \quad (2)$$

where A_L and A_T are the peak areas of linear adsorbed CO and twin-type adsorbed CO, respectively. Obviously, the relative intensity of the linear adsorbed CO (R_L) decreased, while an increased trend was observed for the twin-type adsorbed CO (R_T) on increasing the ΔEN . The twin-type CO occupied a considerable part of the surface sites for the cases of K-Ru/SiO₂, Rb-Ru/SiO₂, and Cs-Ru/SiO₂.

Additionally, if the carbon–oxygen bond in CO was significantly weakened due to electron backdonation, a decreased CH₄ formation temperature should be observed with the decreased electronegativity of the alkali metal promoters in the CO-TPSR profiles.³⁸ However, the CH₄ formation temperature gradually increased with the decrease of alkali metal electronegativity (Figure S7). It seemed that the alkali metal promoters hindered the reactivity of chemisorbed H₂. As described above, the TOF decreased with the addition of the alkali metal promoter with lower electronegativity, since the enhanced electron density on the metallic Ru supported the strong CO adsorption, increased the proportion of inactive twin-type CO, and inhibited the reactivity of hydrogen. Moreover, Figure 10 shows Arrhenius plots with apparent activation

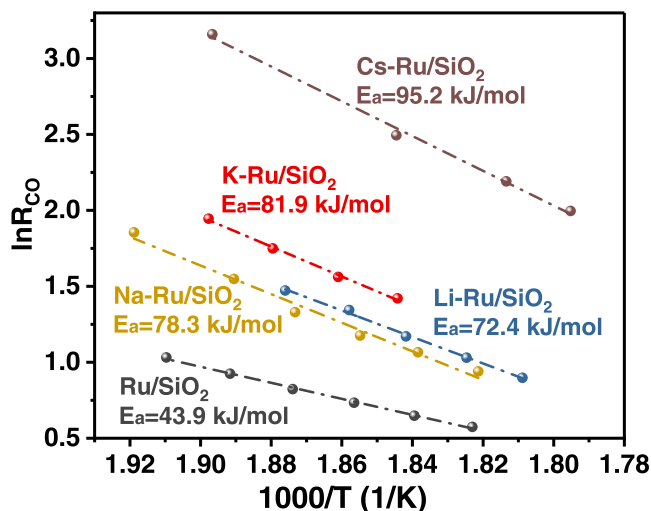


Figure 10. Arrhenius plots of the as-prepared catalysts.

energies (E_a) for AM-Ru/SiO₂. The calculated value of E_a followed the trend Cs-Ru/SiO₂ > K-Ru/SiO₂ > Na-Ru/SiO₂ > Li-Ru/SiO₂ > Ru/SiO₂, which is constant with the catalytic activity variation.³⁹

3. CONCLUSIONS

In this study, we successfully achieved controllable production of long-chain α -olefins by adjusting the type of alkali metal

promoters. The Allen scale electronegativity of the alkali metal promoters could be used as a performance descriptor to clarify the effect of alkali metal promoters on the catalytic performance of Ru-based FTO. Alkali metal promoters with lower electronegativity are more conducive to the adsorption of inactive twin-type CO and inhibit H₂ chemisorption, thus reducing the catalytic activity. The increasing charge density of metallic Ru promotes olefins' desorption and restrains the secondary hydrogenation of olefins, which is beneficial to olefin production, especially for long-chain olefins. Therefore, we provide a common strategy for catalyst design to adjust the product distribution toward long-chain α -olefins with high selectivity from syngas conversion.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.2c06158>.

Experimental section; additional catalytic performance; TEM images; signals of XPS, in situ CO-DRIFTS, and CO-TPSR; and EXAFS fitting parameters (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Liangshu Zhong — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China; School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, P. R. China; orcid.org/0000-0002-4167-8630; Email: zhongls@sari.ac.cn

Authors

Hailing Yu — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China; University of the Chinese Academy of Sciences, Beijing 100049, P. R. China

Yao Wei — University of the Chinese Academy of Sciences, Beijing 100049, P. R. China; Shanghai Institute of Applied Physics, Chinese Academy of Sciences, Shanghai 201800, P. R. China

Tiejun Lin — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China

Caiqi Wang — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China

Yunlei An — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China

Fei Yu — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China

Fanfei Sun — Shanghai Synchrotron Radiation Facility, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China

Zheng Jiang — Shanghai Synchrotron Radiation Facility, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China

Yuhan Sun — CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P. R. China; School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, P. R. China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acscatal.2c06158>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was financially supported by the Natural Science Foundation of China (U22B20136, 22293023, 91945301, 22072177, 22172188), National Key R&D Program of China (2021YFF0500702), Natural Science Foundation of Shanghai (22JC1404200, 21ZR1471700), Program of Shanghai Academic/Technology Research Leader (20XD1404000), the “Transformational Technologies for Clean Energy and Demonstration”, Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDA21020600), and Youth Innovation Promotion Association of CAS. The authors especially acknowledge the XAFS station (BL14W1) of the Shanghai Synchrotron Radiation Facility for the XAS test.

REFERENCES

- (1) Corma, A. Inorganic Solid Acids and Their Use in Acid-Catalyzed Hydrocarbon Reactions. *Chem. Rev.* **1995**, *95*, 559–614.
- (2) Li, Z.; Wu, W.; Wang, M.; Wang, Y.; Ma, X.; Luo, L.; Chen, Y.; Fan, K.; Pan, Y.; Li, H.; Zeng, J. Ambient-pressure hydrogenation of CO₂ into long-chain olefins. *Nat. Commun.* **2022**, *13*, No. 2396.
- (3) Wang, J.; Xu, Y.; Ma, G.; Lin, J.; Wang, H.; Zhang, C.; Ding, M. Directly converting syngas to linear α -olefins over core-shell Fe₃O₄@MnO₂ catalysts. *ACS Appl. Mater. Interfaces* **2018**, *10*, 43578–43587.
- (4) Keim, W. Oligomerization of ethylene to α -olefins: discovery and development of the shell higher olefin process (SHOP). *Angew. Chem., Int. Ed.* **2013**, *52*, 12492–12496.
- (5) Gollwitzer, A.; Dietel, T.; Kretschmer, W. P.; Kempe, R. A broadly tunable synthesis of linear α -olefins. *Nat. Commun.* **2017**, *8*, No. 1226.
- (6) Cheng, K.; Kang, J. C.; King, D. L.; Subramanian, V.; Zhou, C.; Zhang, Q. H.; Wang, Y. Advances in catalysis for syngas conversion to hydrocarbons. *Adv. Catal.* **2017**, *60*, 125–208.
- (7) Lin, T.; An, Y.; Yu, F.; Gong, K.; Yu, H.; Wang, C.; Sun, Y.; Zhong, L. Advances in selectivity control for Fischer–Tropsch synthesis to fuels and chemicals with high carbon efficiency. *ACS Catal.* **2022**, *12*, 12092–12112.
- (8) Torres Galvis, H. M.; Bitter, J. H.; Khare, C. B.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. Supported iron nanoparticles as catalysts for sustainable production of lower olefins. *Science* **2012**, *335*, 835–838.
- (9) Zhong, L.; Yu, F.; An, Y.; Zhao, Y.; Sun, Y.; Li, Z.; Lin, T.; Lin, Y.; Qi, X.; Dai, Y.; Gu, L.; Hu, J.; Jin, S.; Shen, Q.; Wang, H. Cobalt carbide nanoprisms for direct production of lower olefins from syngas. *Nature* **2016**, *538*, 84–87.
- (10) Schulz, H.; Claeys, M. Kinetic modelling of Fischer–Tropsch product distributions. *Appl. Catal., A* **1999**, *186*, 91–107.
- (11) Van Der Laan, G. P.; Beenackers, A. A. C. M. Kinetics and Selectivity of the Fischer–Tropsch Synthesis: A Literature Review. *Catal. Rev.* **1999**, *41*, 255–318.
- (12) Zhai, P.; Xu, C.; Gao, R.; Liu, X.; Li, M.; Li, W.; Fu, X.; Jia, C.; Xie, J.; Zhao, M.; Wang, X.; Li, Y. W.; Zhang, Q.; Wen, X. D.; Ma, D. Highly tunable selectivity for syngas-derived alkenes over zinc and sodium-modulated Fe₃C₂ catalyst. *Angew. Chem., Int. Ed.* **2016**, *55*, 9902–9907.
- (13) Yu, X.; Zhang, J.; Wang, X.; Ma, Q.; Gao, X.; Xia, H.; Lai, X.; Fan, S.; Zhao, T.-S. Fischer–Tropsch synthesis over methyl modified Fe₂O₃@SiO₂ catalysts with low CO₂ selectivity. *Appl. Catal., B* **2018**, *232*, 420–428.
- (14) Liu, Y.; Xu, M.; Yang, Z.; Ding, X.; Zhu, M.; Han, Y.-F. Syngas to olefins with low CO₂ formation by tuning the structure of Fe_{C_x}-MgO-Al₂O₃ catalysts. *Chem. Eng. J.* **2022**, *450*, No. 137167.
- (15) Lin, T.; Liu, P.; Gong, K.; An, Y.; Yu, F.; Wang, X.; Zhong, L.; Sun, Y. Designing silica-coated CoMn-based catalyst for Fischer–Tropsch synthesis to olefins with low CO₂ emission. *Appl. Catal., B* **2021**, *299*, No. 120683.
- (16) Xu, Y.; Li, X.; Gao, J.; Wang, J.; Ma, G.; Wen, X.; Yang, Y.; Li, Y.; Ding, M. A hydrophobic FeMn@Si catalyst increases olefins from syngas by suppressing C1 by-products. *Science* **2021**, *371*, 610–613.
- (17) Gong, K.; Wei, Y.; Dai, Y.; Lin, T.; Yu, F.; An, Y.; Wang, X.; Sun, F.; Jiang, Z.; Zhong, L. Carbon-encapsulated metallic Co nanoparticles for Fischer–Tropsch to olefins with low CO₂ selectivity. *Appl. Catal., B* **2022**, *316*, No. 121700.
- (18) Jeske, K.; Kizilkaya, A. C.; Lopez-Luque, I.; Pfaender, N.; Bartsch, M.; Concepcion, P.; Prieto, G. Design of cobalt Fischer–Tropsch catalysts for the combined production of liquid fuels and olefin chemicals from hydrogen-rich syngas. *ACS Catal.* **2021**, *11*, 4784–4798.
- (19) Xiang, Y.; Kruse, N. Tuning the catalytic CO hydrogenation to straight- and long-chain aldehydes/alcohols and olefins/paraffins. *Nat. Commun.* **2016**, *7*, No. 13058.
- (20) Cheng, Y.; Lin, J.; Xu, K.; Wang, H.; Yao, X.; Pei, Y.; Yan, S.; Qiao, M.; Zong, B. Fischer–Tropsch synthesis to lower olefins over potassium-promoted reduced graphene oxide supported iron catalysts. *ACS Catal.* **2016**, *6*, 389–399.
- (21) Luo, M.; O'Brien, R. J.; Bao, S.; Davis, B. H. Fischer–Tropsch synthesis: induction and steady-state activity of high- α potassium promoted iron catalysts. *Appl. Catal., A* **2003**, *239*, 111–120.
- (22) Panagiotopoulou, P. Methanation of CO₂ over alkali-promoted Ru/TiO₂ catalysts: II. Effect of alkali additives on the reaction pathway. *Appl. Catal., B* **2018**, *236*, 162–170.
- (23) Lillebø, A. H.; Patanou, E.; Yang, J.; Blekkan, E. A.; Holmen, A. The effect of alkali and alkaline earth elements on cobalt based Fischer–Tropsch catalysts. *Catal. Today* **2013**, *215*, 60–66.
- (24) Torres Galvis, H. M.; Koeken, A. C. J.; Bitter, J. H.; Davidian, T.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. Effects of sodium and sulfur on catalytic performance of supported iron catalysts for the Fischer–Tropsch synthesis of lower olefins. *J. Catal.* **2013**, *303*, 22–30.
- (25) Xie, J.; Paalanen, P. P.; van Deelen, T. W.; Weckhuysen, B. M.; Louwerse, M. J.; de Jong, K. P. Promoted cobalt metal catalysts suitable for the production of lower olefins from natural gas. *Nat. Commun.* **2019**, *10*, No. 167.
- (26) Li, Z.; Zhong, L.; Yu, F.; An, Y.; Dai, Y.; Yang, Y.; Lin, T.; Li, S.; Wang, H.; Gao, P.; Sun, Y.; He, M. Effects of sodium on the catalytic performance of CoMn catalysts for Fischer–Tropsch to olefin reactions. *ACS Catal.* **2017**, *7*, 3622–3631.
- (27) Li, Z.; Yao, N.; Cen, J.; Li, X.; Zhong, L.; Sun, Y.; He, M. Effects of alkali metal promoters on the structure–performance relationship of CoMn catalysts for Fischer–Tropsch synthesis. *Catal. Sci. Technol.* **2020**, *10*, 1816–1826.
- (28) Yu, H.; Wang, C.; Lin, T.; An, Y.; Wang, Y.; Chang, Q.; Yu, F.; Wei, Y.; Sun, F.; Jiang, Z.; Li, S.; Sun, Y.; Zhong, L. Direct production of olefins from syngas with ultrahigh carbon efficiency. *Nat. Commun.* **2022**, *13*, No. 5987.
- (29) Zhang, Y.; Su, X.; Li, L.; Qi, H.; Yang, C.; Liu, W.; Pan, X.; Liu, X.; Yang, X.; Huang, Y.; Zhang, T. Ru/TiO₂ catalysts with size-dependent metal/support interaction for tunable reactivity in Fischer–Tropsch synthesis. *ACS Catal.* **2020**, *10*, 12967–12975.
- (30) Okuhara, T.; Tomoo, K.; Kenji, K.; Makoto, M.; Yukio, Y. Effects of Dispersion in Carbon Monoxide Adsorption and Carbon Monoxide Hydrogenation over Alumina-supported Ruthenium Catalysts. *Bull. Chem. Soc. Jpn.* **1984**, *57*, 938–943.

- (31) Okuhara, T.; Tamura, H.; Misono, M. Effect of potassium and phosphorus on the hydrogenation of CO over alumina-supported ruthenium catalyst. *J. Catal.* **1985**, *95*, 41–48.
- (32) Ramamoorthy, P.; Gonzalez, R. D. Surface characterization of supported Pt-Ru bimetallic clusters using infrared spectroscopy. *J. Catal.* **1979**, *58*, 188–197.
- (33) Mazziere, V.; Coloma-Pascual, F.; Arcoya, A.; L'Argentière, P. C. XPS, FTIR and TPR characterization of Ru/Al₂O₃ catalysts. *Appl. Surf. Sci.* **2003**, *210*, 222–230.
- (34) Zhang, Y.; Yang, X.; Yang, X.; Duan, H.; Qi, H.; Su, Y.; Liang, B.; Tao, H.; Liu, B.; Chen, D.; Su, X.; Huang, Y.; Zhang, T. Tuning reactivity of Fischer-Tropsch synthesis by regulating TiO_x overlayer over Ru/TiO₂ nanocatalysts. *Nat. Commun.* **2020**, *11*, No. 3185.
- (35) Chen, W.; Cao, J.; Yang, J.; Cao, Y.; Zhang, H.; Jiang, Z.; Zhang, J.; Qian, G.; Zhou, X.; Chen, D.; Yuan, W.; Duan, X. Molecular-level insights into the electronic effects in platinum-catalyzed carbon monoxide oxidation. *Nat. Commun.* **2021**, *12*, No. 6888.
- (36) Yang, J.; Ma, W.; Chen, D.; Holmen, A.; Davis, B. H. Fischer–Tropsch synthesis: A review of the effect of CO conversion on methane selectivity. *Appl. Catal., A* **2014**, *470*, 250–260.
- (37) Loveless, B. T.; Buda, C.; Neurock, M.; Iglesia, E. CO chemisorption and dissociation at high coverages during CO hydrogenation on Ru catalysts. *J. Am. Chem. Soc.* **2013**, *135*, 6107–6121.
- (38) Sakakini, B. H.; S Verbrugge, A. Temperature-programmed surface reaction as a means of characterizing supported-metal catalysts and probing their surface reactivity. *J. Chem. Soc., Faraday Trans.* **1997**, *93*, 1637–1640.
- (39) Lyu, S.; Cheng, Q.; Liu, Y.; Tian, Y.; Ding, T.; Jiang, Z.; Zhang, J.; Gao, F.; Dong, L.; Bao, J.; Ma, Q.; Yang, Q.-H.; Li, X. Dopamine sacrificial coating strategy driving formation of highly active surface-exposed Ru sites on Ru/TiO₂ catalysts in Fischer–Tropsch synthesis. *Appl. Catal., B* **2020**, *278*, No. 119261.

Identifying Performance Descriptor in Direct Syngas Conversion to Long-chain α -olefins over Ruthenium-based Catalysts Promoted by Alkali Metals

Hailing Yu^{1,2}, Yao Wei^{2,4}, Tiejun Lin¹, Caiqi Wang¹, Yunlei An¹, Fei Yu¹, Fanfei Sun⁵,
Zheng Jiang⁵, Yuhan Sun^{1,3}, Liangshu Zhong^{1,3*}

¹ CAS Key Laboratory of Low-Carbon Conversion Science and Engineering, Shanghai
Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, P.R. China

² University of the Chinese Academy of Sciences, Beijing 100049, P.R. China

³ School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210,
P.R. China

⁴ Shanghai Institute of Applied Physics, Chinese Academy of Sciences, Shanghai 201800,
P.R. China

⁵ Shanghai Synchrotron Radiation Facility, Shanghai Advanced Research Institute, Chinese
Academy of Sciences, Shanghai 201210, P.R. China

*Corresponding author. Email: zhongls@sari.ac.cn

Table of Content

Experimental Section	3
Chemicals	3
Catalyst Preparation	3
Catalyst Characterization	3
Catalytic Evaluation	5
Figure S1. Representative TEM images and the corresponding crystallize size distributions for various spent catalysts.	8
Figure S2. XPS Ru 3 <i>d</i> spectra of various AM-Ru/SiO ₂ catalysts after reduction by H ₂ at 450 °C for 4 h.	9
Figure S3. Effect of alkali metal promoter on the olefin-to-paraffin ratio of hydrocarbons.	10
Figure S4. ASF distribution of olefins products for various xK-Ru/SiO ₂ catalysts at 270 °C, 5 bar, 3000 mL·g _{cat} ⁻¹ ·h ⁻¹ and H ₂ /CO=2.	11
Figure S5. Generation of CH _x species during H ₂ flow at 270 °C as determined using in situ CO-DRIFTS over (a) Li-Ru/SiO ₂ , (b) K-Ru/SiO ₂ and (c) Cs-Ru/SiO ₂	12
Figure S6. CO-DRIFTS spectra with partial enlarged detail at the range of 2100 –1850 cm ⁻¹ over AM-Ru/SiO ₂ catalysts.	13
Figure S7. CO-TPSR profiles for AM-Ru/SiO ₂ catalysts.	14
Table S1. EXAFS fitting parameters at the Ru-edge (S ₀ ² =0.75).	15
Table S2. Chemisorption results of various AM-Ru/SiO ₂ catalysts.	16

Experimental Section

Chemicals

Ruthenium nitrosyl nitrate solution (0.1414 g Ru per gram of solution) was purchased from Heraeus Precious Metal Technology Co., Ltd. Aerosol silica (SiO_2 , AEROSIL 380) was purchased from Evonik Degussa China Co., Ltd. Alkali metal nitrates including LiNO_3 , NaNO_3 , KNO_3 , RbNO_3 and CsNO_3 (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd. All materials were used as received without further purification.

Catalyst Preparation

All alkali metal promoted Ru/ SiO_2 catalysts were prepared via incipient wetness impregnation method. The mixture solution of ruthenium nitrosyl nitrate and alkali metal nitrate (molar ratio of alkali metal to Ru = 0.5) was used for the incipient wetness impregnation of aerosol silica (SiO_2). The as-prepared samples denoted as AM-Ru/ SiO_2 (AM = Li, Na, K, Rb, Cs) were dried at 60 °C for 4 h and calcinated in air at 400 °C for 3 h with a ramping rate of 2 °C/min. ICP-OES and XRF results showed the analogical Ru loading (4.0 wt.%) and AM/Ru molar ratio (0.5) for each sample. The promoted catalysts with different K content were prepared via similar procedure with different amount of KNO_3 . The samples were denoted as xK-Ru/ SiO_2 (x=0.1, 0.3, 0.5 and 1.0), where x represents the K/Ru molar ratio.

Catalyst Characterization

The elemental composition of various samples was measured by inductively coupled plasma optical emission spectrometry (ICP-OES, Perkin Elmer) and X-ray fluorescence (XRF, Rigaku ZSX Primus II).

N_2 physisorption experiments were carried out on Micromeritics 2020 instrument at -196 °C. Brunauer-Emmett-Tellwer (BET) method was used to

calculate the specific surface areas. Pore volume and pore size were determined by the Barrett-Joyner-Halenda (BJH) method.

Power X-ray diffraction (XRD) was performed on a Rigaku Ultima IV X-ray diffractometer (40 kV, 40 mA) equipped with Cu K α radiation ($\lambda = 1.54056 \text{ \AA}$), which was operated at 40 kV and 40 mA with a scanning rate of 2 degree/min.

Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) measurements were conducted on a FEI Tecnai G2 F20 S-TWIN equipment with 200 kV accelerating voltage. The nanoparticle size distribution for each sample was determined using at least 200 nanoparticles.

X-ray photoelectron spectroscopy (XPS) data was collected by Thermo Fisher Scientific K-Alpha spectrometer equipped with the Al K α (1486.6 eV) radiation source. The results were calibrated with C1s peak at 284.8 eV.

H₂ temperature-programmed reduction (H₂-TPR) was performed on Micromeritics ChemiSorb2920 instrument with a thermal conductivity detector (TCD). Before TPR measurement, the samples were pretreated in He flow at 120 °C for 60 min, then cooled down to 50 °C. The reactor was heated to 800 °C at a heating rate of 10 °C·min⁻¹ in 5 vol% H₂ (balance Ar, 30 mL·min⁻¹).

H₂ and CO chemisorption experiments were conducted with a Micromeritics 2020 instrument. Typically, 100 to 150 mg of sample was degassed at 300 °C for 4 h. Prior to the chemisorption experiments, the sample was in situ treated at 450 °C in H₂ for 2 h and then evacuated. The temperature was decreased to 40 °C under vacuum, and then was evacuated for 60 min. The adsorbate (H₂ or CO) was introduced into system to record the chemisorption isotherms. In order to compare the effect of adsorbed CO on H₂ adsorption, H₂ chemisorption was also measured over the reduced sample after CO pre-adsorption and evacuation at 40 °C.

In situ diffuse reflectance infrared Fourier transform spectra of CO adsorption (CO-DRIFTS) were conducted on ThermoFisher Scientific FTIR spectrometer (Nicolet iS50). Prior to CO adsorption, the sample was pretreated

in H₂ flow at 450 °C for 1 h and then the gas flow switched to Ar flow to purge and cooled down to 50 °C or 270 °C. The background spectra were collected at 50 °C and 270 °C under Ar flow. After that, the CO flow was introduced until saturated adsorption. The CO-DRIFTS spectra were obtained until no gas-phase CO with Ar purging. As for the adsorbed CO reactivity experiment at 270 °C, a flow of diluted H₂ was introduced into the reactors and the spectra were automatically recorded to acquire the reactivity of adsorbed CO.

Temperature-programmed surface reaction (TPSR) of CO was carried out in a Micromeritics Chemsorb2920 with a thermal conductivity detector (TCD) and MKS Cirrus 2 mass spectrometer (MS). The sample was pretreated in a H₂ flow of H₂ at 450 °C for 2 h and then flushing with He for 30 min. The temperature was cooled down to 50 °C and CO flow was switched to realize the saturated CO adsorption. The sample was heated up to 500 °C at 10 °C·min⁻¹ in flowing of H₂ flow, and the signals of m/z=16 (CH₄) was monitored by mass spectrometer.

X-ray absorption spectroscopy (XAS) experiments were conducted on the BL11B beamline of Shanghai Synchrotron Radiation Facility (SSRF), China. The Ru K-edge spectra were collected at room temperature with transition mode.

Catalytic Evaluation

Syngas conversion over various catalysts was carried out in a continuous fixed bed made of stainless steel. Before catalytic testing, 1 g sample (40–60 mesh) was in situ reduced at 450 °C for 4 h in pure H₂ flowing with a heating ramp of 2 °C/min. The feed gas (H₂/CO=2, including 3% N₂ as inner standard) was introduced after cooling down to 180 °C, then the reaction pressure was controlled at 5 bar. The reaction temperature was adjusted to 270 °C with weight hourly space velocity (WHSV) of 3000 mL·g⁻¹·h⁻¹. The outlet gas compositions were analyzed by an on-line gas chromatograph (Agilent 7890B)

equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). The liquid product was collected and analyzed by off-line Shimadzu GC equipped with FID. In order to obtain similar CO conversion level (< 10%) for various catalysts, the evaluation was performed by adjusting reaction space velocity.

CO conversion was calculated according to the following equation:

$$X_{CO} = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} \times 100\% \quad (1)$$

Where CO_{inlet} and CO_{outlet} refer to the molar numbers of CO at the inlet and outlet, respectively.

Product selectivity was calculated according to the following equation:

$$S_i = \frac{N_i \times n_i}{\sum(N_i \times n_i)} \times 100\% \quad (2)$$

where S_i denotes the selectivity to product i on a carbon basis, N_i is the molar fraction of product i, and n_i is the carbon number of product i. The selectivity of oxygenates was less than 1% and has been excluded from the reported product selectivity.

The reaction rate was calculated as following:

$$\text{Reaction rate}_{CO} = \frac{WHSV \times X_{CO} \times CO \text{ Concentration}}{22400 \times \xi_{Ru}} \quad (3)$$

Where WHSV is the weight hourly space velocity, X_{CO} is the CO conversion, CO Concentration represents the molar concentration of CO in the feedstock, ξ_{Ru} represents the loading amount of Ru measured by ICP-AES.

The turnover frequency (TOF) value was calculated as following:

$$TOF = \frac{\text{Reaction rate}_{CO} \times M_{Ru}}{D_{Ru} \times 3600} \quad (4)$$

Where M_{Ru} is the relative atomic mass of Ru (101.07 g/mol), D_{Ru} is the dispersion of metallic Ru measured by CO chemisorption.

The chain growth probability (α) was calculated according to Anderson-Schulz-Flory distribution:

$$\left(\frac{W_n}{n}\right) = (1 - \alpha)^2 \alpha^{(n-1)} \quad (5)$$

where n is the carbon number of products, W_n is mass fraction of the hydrocarbons with a carbon number of n , α is chain growth probability.

The equation of (4) can be edited as:

$$\ln\left(\frac{W_n}{n}\right) = (n - 1)\ln\alpha + 2\ln(1 - \alpha) \quad (6)$$

Plotting $\ln(W_n/n)$ versus n (carbon number), and the chain growth probability (α) can be obtained by calculating the slope ($\ln\alpha$).

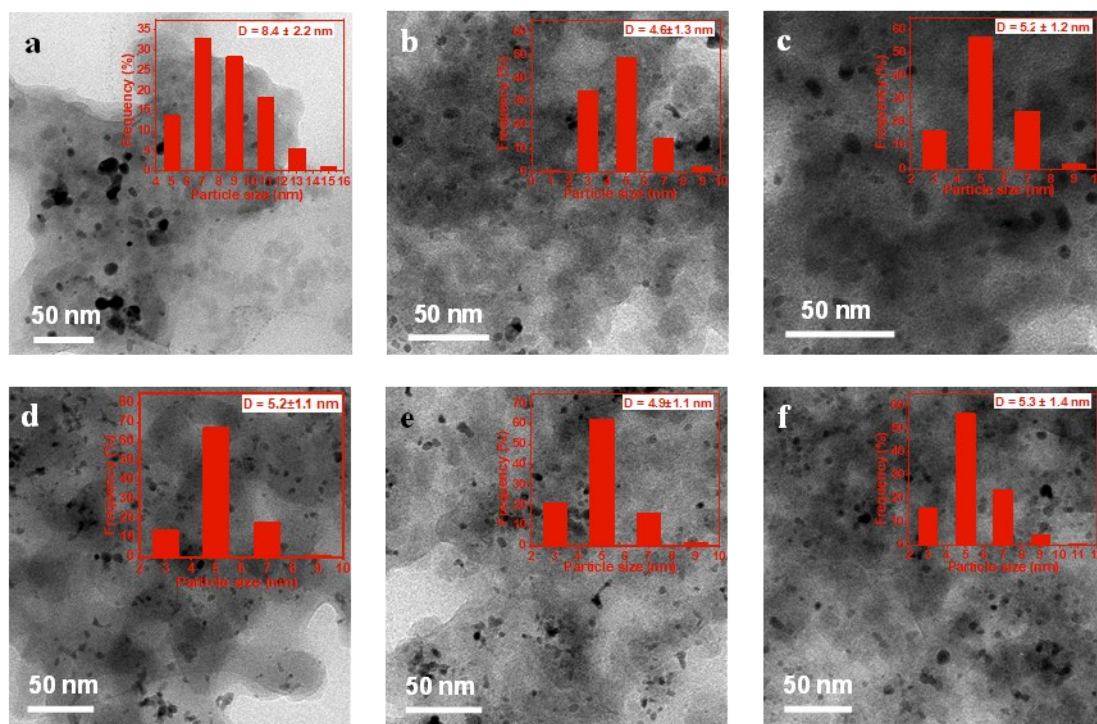


Figure S1. Representative TEM images and the corresponding crystallite size distributions for various spent catalysts. (a) Ru/SiO₂, (b) Li-Ru/SiO₂, (c) Na-Ru/SiO₂, (d) K-Ru/SiO₂, (e) Rb-Ru/SiO₂, (f) Cs-Ru/SiO₂.

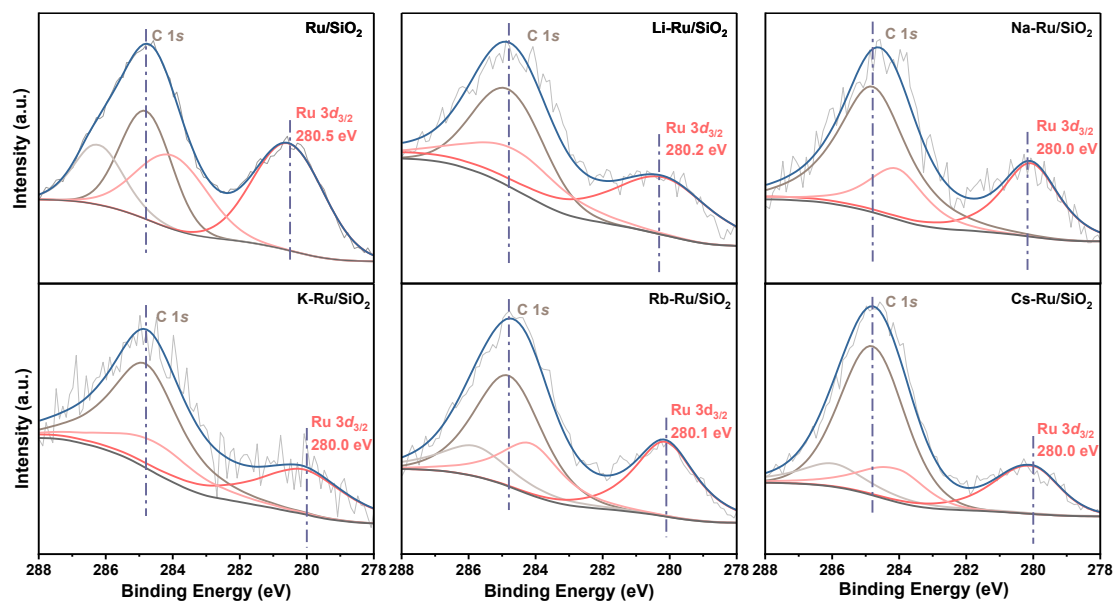


Figure S2. XPS Ru 3d spectra of various AM-Ru/SiO₂ catalysts after reduction by H₂ at 450 °C for 4 h.

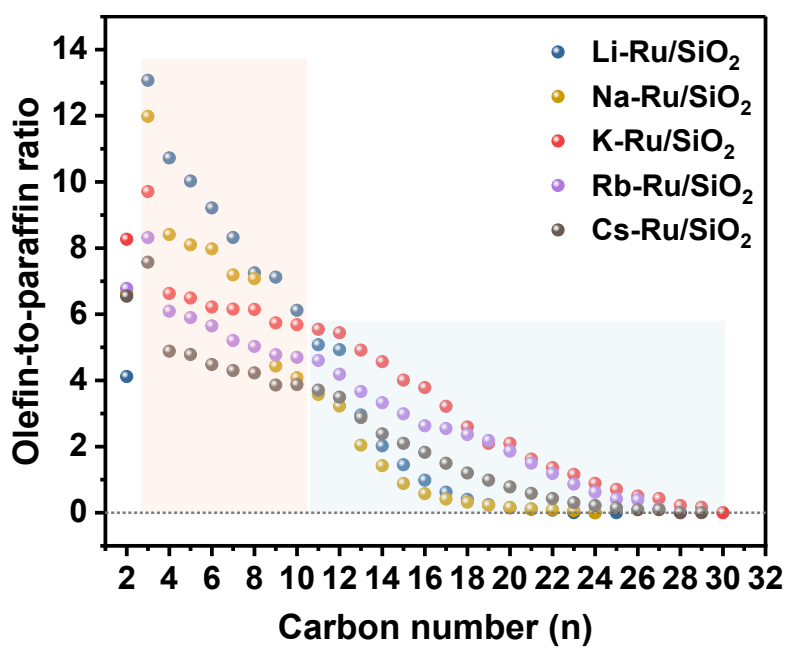


Figure S3. Effect of alkali metal promoter on the olefin-to-paraffin ratio of hydrocarbons.

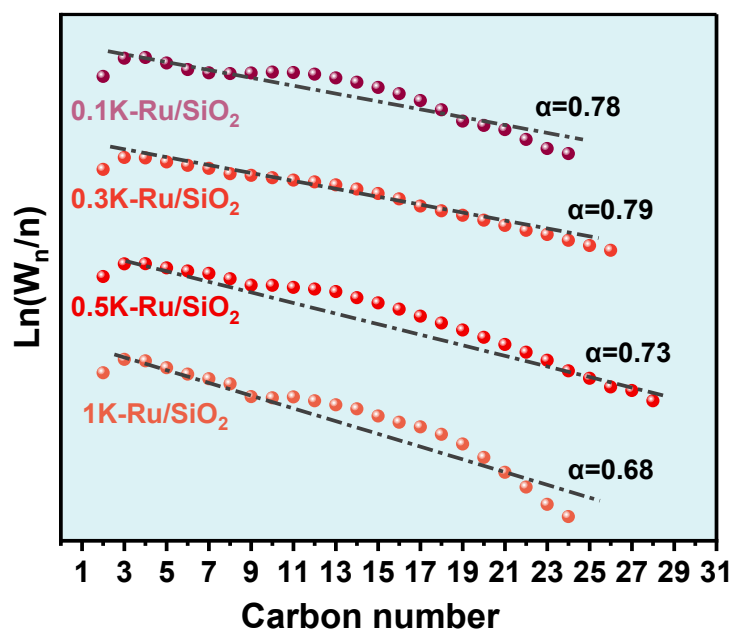


Figure S4. ASF distribution of olefins products for various xK-Ru/SiO₂ catalysts at 270 °C, 5 bar, 3000 mL·g_{cat.}⁻¹·h⁻¹ and H₂/CO=2.

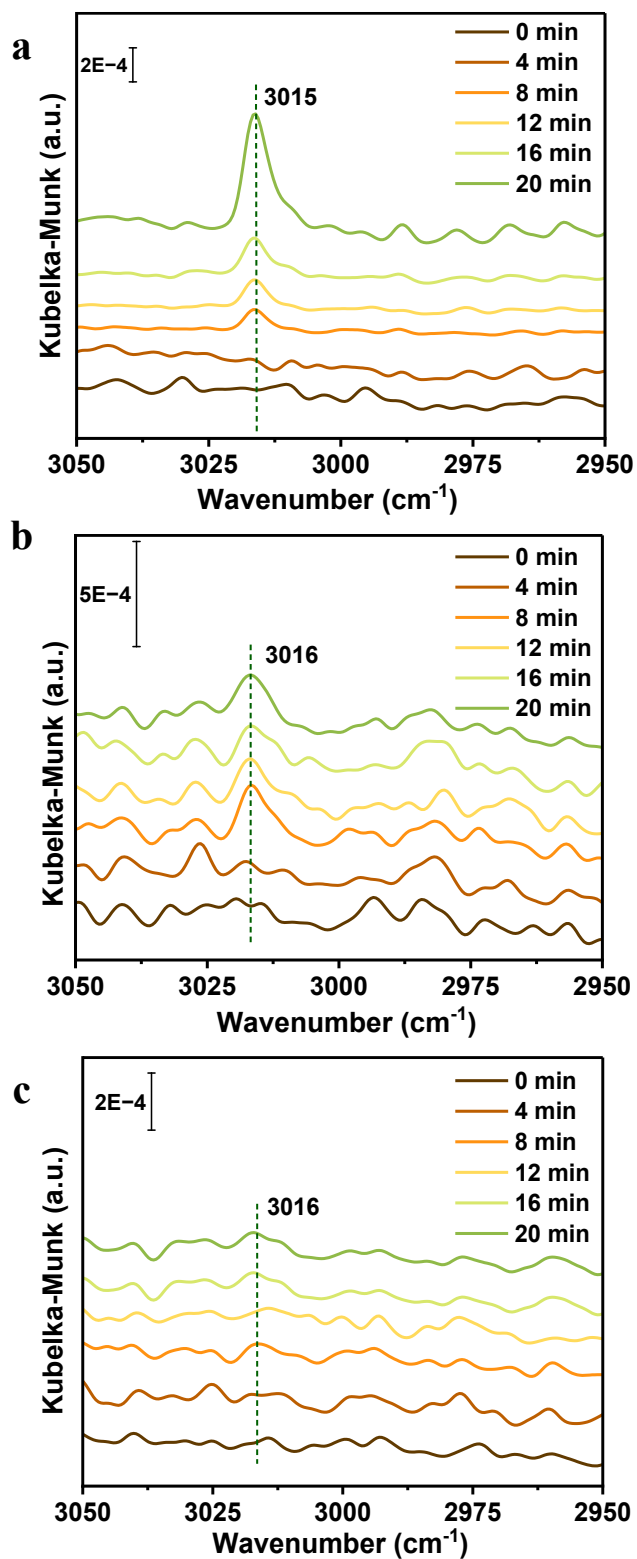


Figure S5. Generation of CH_x species during H₂ flow at 270 °C as determined using in situ CO-DRIFTs over (a) Li-Ru/SiO₂, (b) K-Ru/SiO₂ and (c) Cs-Ru/SiO₂.

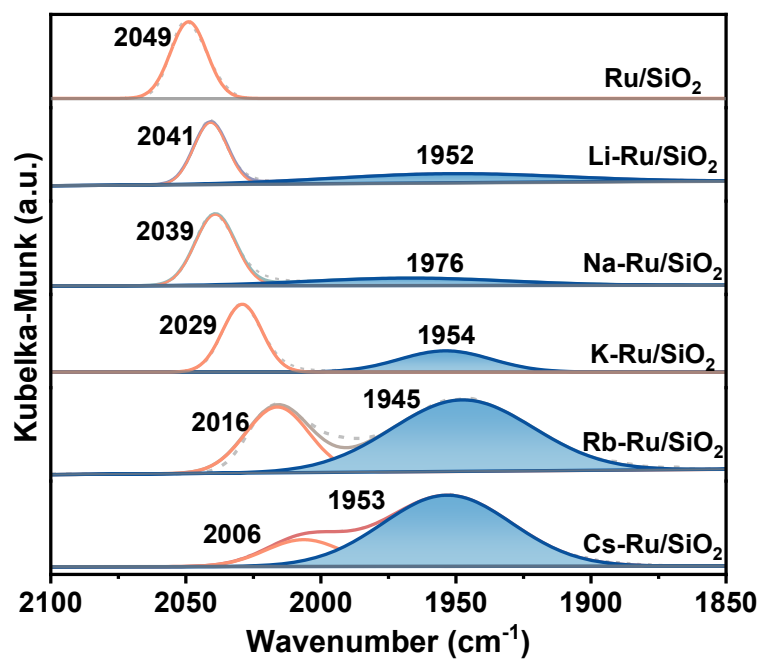


Figure S6. CO-DRIFTS spectra with partial enlarged detail at the range of 2100 –1850 cm⁻¹ over AM-Ru/SiO₂ catalysts.

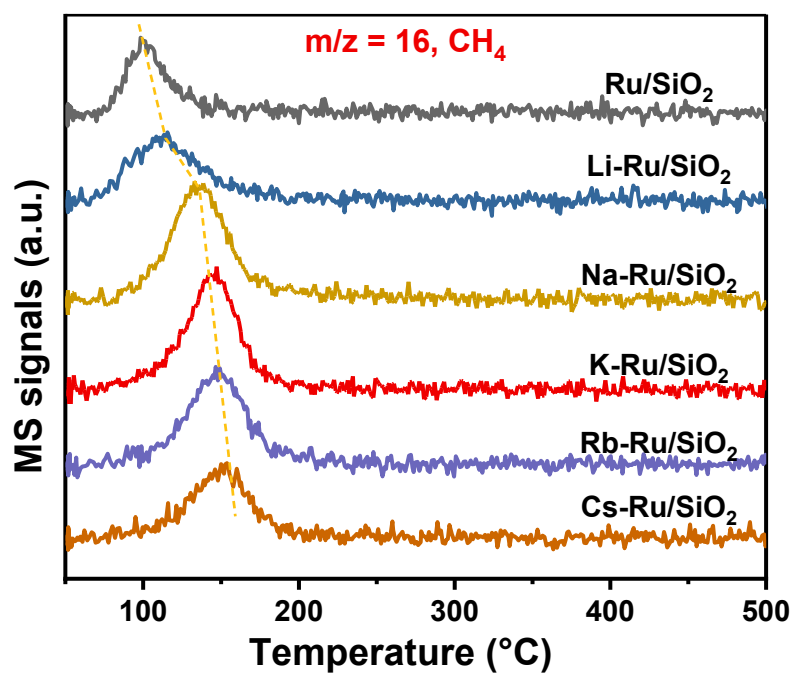


Figure S7. CO-TPSR profiles for AM-Ru/SiO₂ catalysts.

Table S1. EXAFS fitting parameters at the Ru-edge ($S_0^2=0.75$).

Sample	Shell	CN ^a	^b R(Å)	^c $\sigma^2 \times 10^2 (\text{\AA}^2)$	^d $\Delta E_0 (\text{eV})$	R factor
Ru foil	Ru-Ru	12.0	2.71	-	-	-
Li-Ru/SiO ₂	Ru-Ru	10.9	2.67	0.42±0.05	6.27±1.19	0.007
Na-Ru/SiO ₂	Ru-Ru	10.5	2.67	0.46±0.05	5.55±1.06	0.006
K-Ru/SiO ₂	Ru-Ru	10.3	2.67	0.41±0.05	5.47±1.05	0.009
Rb-Ru/SiO ₂	Ru-Ru	10.0	2.67	0.44±0.05	5.05±1.05	0.006
Cs-Ru/SiO ₂	Ru-Ru	10.7	2.67	0.45±0.07	5.66±1.34	0.008
RuO ₂	Ru-O	6.0	1.99	-	-	-

^a CN is the coordination number for the adsorber-backscatter pair.

^b R is the average interatomic distance.

^c σ^2 is the Debye-Waller factor.

^d ΔE_0 is the inner potential correction.

Table S2. Chemisorption results of various AM-Ru/SiO₂ catalysts.

Catalysts	H/Ru	H/Ru ^a	CO/Ru
Li-Ru/SiO ₂	0.14	0.11	0.30
Na-Ru/SiO ₂	0.16	0.09	0.31
K-Ru/SiO ₂	0.26	0.14	0.33
Rb-Ru/SiO ₂	0.20	-	0.31
Cs-Ru/SiO ₂	0.16	0.12	0.30

^a The H/Ru value was calculated after CO pre-adsorbed.